TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A MAJOR PROJECT FINAL REPORT

ON

"EVALUATING THE EFFECTIVENES OF VISVAP-PROGRAMMED SEMI-ACTUATED SIGNAL CONTROL OVER EXISTING FIXED-TIME CONTROL: A CASE STUDY OF BABARMAHAL INTERSECTION, KATHMANDU VALLEY "

Submitted By:

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Submitted To:

Department of Civil Engineering

Thapathali Campus

Kathmandu, Nepal

Under the Supervision of

Dr. Rojee Pradhananga

February 2025

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ABSTRACT

150-300 words, include: Problem context, Approach, Key results, Impact. Avoid citations, unexpanded acronyms, detailed implementation.

Keywords—Combinatorial optimization, constraint satisfaction, genetic algorithm, hyper-heuristics, metaheuristic search, NP-hard problems, reinforcement learning, university course timetabling problem.

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1. INTRODUCTION

1.1. Background

Urban Traffic Congestion at signalized Intersection remains a Critical challenge in modern transportation system. Traffic congestion is the degradation of traffic flow condition on road and become slower speed which increase the travel time and increased Vehicle queues. It occurs when the demand of road space exceeds its capacity. Intersection Congestion is the main problem in urban area occurs when vehicle demand exceeds the capacity of a signalized or unsignalized crossing, leading to delays, queuing, and reduced traffic flow efficiency. Unlike general intersection congestion is often caused by conflicting vehicle movements, inefficient signal timing.

Congestion arises when there are more vehicles on the road than the infrastructure can handle. Significant delays in urban areas frequently originate at intersections, where conflicting traffic streams converge. There is extensive experience with chronic gridlock and congestion in the Kathmandu Valley. (Tiwari et al., 2023).

One of the popular concepts in traffic signalization is signalization is Fixed time traffic signal. A Fixed- time traffic signal is a traditional signal control system that operates on predetermined, unchanging timing plans. These signal cycle through preset green, yellow, and red phase at regular interval, regardless of real traffic demand.

Fixed time Traffic signal is wastes of times on empty approaches, which increases the delays in peak hours. When traffic demand fluctuated, the efficacy of Optimum Fixed-Time Management declined, resulting in significant increases in average vehicle delay, fuel consumption, and exhaust emissions. To solve the wastes of times on empty approaches, we introduce the actuated traffic signal. An actuated traffic signal is a signal control system that adjusts its timing in real-time based on vehicle or pedestrian demand, detected through sensor (e.g. inductive loops, camera, and radar).

A semi-Actuated Traffic signal is a type of Traffic Control system that combines elements of both fixed-time and fully actuated signals. It is mostly designed for the intersection where the major road carries a significantly higher traffic volume most of the time, while intersecting road i.e. minor road have experience lower, variable traffic flow. Semi-actuated signals operate with vehicle detection only on the minor approaches. The major road usually receives a consistent green phase by default. Vehicle arriving on the minor road are detected by sensors (like inductive loops or cameras).

Microsimulation tools like PTV VISSIM with VISVAP are widely used to modal and optimize actuated signal control. VISSIM is best software to simulation capabilities enable precise evaluation of traffic dynamics, while VISVAP facilitates the programming of adaptive signal logic. VISSIM simulation program, the time intervals when the fluctuations in traffic demands occur and the quantities (proportions) of the fluctuations at these time intervals can be determined by the users. Thus, changes in intersection performance resulting from short- or long-term fluctuations caused by concerts, sports events, traffic accidents, or adverse weather can be analyzed using VISSIM. (Cakici et al., 2021) .

1.2. Research Gaps and Problem Statement

Traffic congestion at major intersections throughout the Kathmandu Valley has reached critical levels, significantly degrading urban mobility, the economy, and public well-being. The current situation at the Babarmahal intersection is at a critical phase. The current fixed-time traffic signal system operates inefficiently, failing to adapt to the highly asymmetric traffic flows observed at this junction. The main issue of this intersection is the mismatched allocation of green signal time relative to actual Vehicle demand. The two major legs carry a high traffic volume, i.e., Baneshowr – Babarmahal leg and Maitighar-Babarmahal, whereas the two minor legs, i.e., toward Department of Road and toward Arts Gallery, carry a low traffic Volume. The inefficient Traffic light signal, i.e., long green time interval to the minor legs during periods when these minor legs experienced little or no Vehicular flow. Valuable green time is wasted on this time, which leads to heavily congested major roads and increases delay time.

The absence of the demand-responsive signal control is the main problem of this intersection. While a semi-actuated traffic signal system, which dynamically adjusts minor leg green time based on real-time vehicle detection , i.e., changes the traffic light toward the minor leg and gives the green on the major leg while there are no vehicles on the minor leg.

1.3. Objectives

This study aims to:

1. Assess the performance of a four-legged signalized intersection under existing fixed-time control using PTV VISSIM.
2. Optimize signal timings via actuated control in VISVAP, focusing on parameters like detector placement and green time allocation.
3. Compare the results with fixed-time control to quantify improvements in delay and queue length.

1.3.1. Study Area

The study focuses on the critical Babarmahal Road intersection (27.692506N,85.323863E) located in the administrative and cultural heart of Kathmandu, Nepal. It is a four-legged signalized intersection. The intersection features four distinct legs:

a. Major East leg: Carry a high-Volume Traffic along the Baneshwor-Babarmahal axis.
b. Major West leg: Carry a high-volume traffic along the Maitighar- Babarmahal axis.
c. Minor North leg: Leading directly towards the Department of Road (DOR).
d. Minor south leg: leading towards the Arts Gallery.

Figure 1-1. Aerial view of Intersection from Google Earth Pro

Figure 1-2. Intersection during visit

The traffic volume along the Maitighar (central) approach and the Baneshwor approach is significantly higher, classifying them as major roads within the intersection. In contrast, the Arts Gallery approach and the Department of Roads approach experience relatively lower and more fluctuating traffic volumes throughout the day, making them minor roads. Given this variation in traffic demand, implementing a semi-actuated traffic signal control system at this intersection is expected to be beneficial. Such a system can help reduce queue lengths, minimize vehicle delays, improve the Level of Service (LOS), and offer other operational advantages.

Geometric Specifications of Babarmahal Intersection

Table 1-1. Geometrical Parameters

Parameters	**Specifications**
Number of Legs	4(Four-legged intersection)
Approaches	Maitighar, Baneshwor, Arts Gallery, Department of Road (DOR)
Number of lanes Per Approach	Maitighar Approach: 8 lanes Baneshwor Approach: 8 lanes ArtsGallery Approach: 2 lanes DOR Approach: 2 lanes
Lane Width	**Maitighar Approach:** Service Lane: 3.5m(Total 4 lanes) Main Lane: 3m(Total 4 lanes) **Baneshowr Approach:** Service Lane: 3.5m(Total 4 lanes) Main Lane: 3m(Total 4 lanes) **ArtsGallery Approach:** Main Lane: about 2.5m(Total 2 lanes) **DOR Approach:** Main Lane: 2.5m(Total 2 lanes)
Pedestrian Crosswalks	Present on two legs (Maitighar approach and Arts Gallery approach)
Crosswalk Length	About 3.1m
Existing Signal Poles	Present at all four corners (Fixed-Time Signal Control)

NOTE: The above measurements are preliminary estimates obtained using Google Earth Pro and will be verified through field surveys.

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